

## Calibration Lab Vetting Guide

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How to verify accreditation, evaluate measurement capability, and avoid unqualified calibration providers

- ✓ ISO/IEC 17025 accreditation, A2LA, NVLAP, and ANAB explained
- ✓ Step-by-step accreditation verification process
- ✓ 10 red flags that indicate an unqualified calibration lab
- ✓ Sample vetting questionnaire you can send to labs

# Understanding Calibration Lab Accreditation

ISO/IEC 17025:2017 is the international standard for the competence of testing and calibration laboratories. Accreditation to this standard — verified by an independent accreditation body — is the single most important credential a calibration lab can hold. It demonstrates that the lab has been independently assessed for technical competence, has a documented quality management system, and participates in proficiency testing to verify measurement accuracy.

## Accreditation Framework

| Accreditation Body | Full Name                                           | Focus                                 | Recognized By                        |
|--------------------|-----------------------------------------------------|---------------------------------------|--------------------------------------|
| A2LA               | American Association for Laboratory Accreditation   | Largest US calibration lab accreditor | ILAC MRA (global mutual recognition) |
| NVLAP              | National Voluntary Laboratory Accreditation Program | NIST-administered accreditation       | US federal agencies, ILAC MRA        |
| ANAB               | ANSI National Accreditation Board                   | Broad accreditation services          | ILAC MRA                             |
| PJLA               | Perry Johnson Laboratory Accreditation              | Growing US accreditor                 | ILAC MRA                             |
| IAS                | International Accreditation Service                 | Building/construction focus           | ILAC MRA                             |

## What ISO/IEC 17025:2017 Requires

| Requirement                          | What It Means for You                                                                        |
|--------------------------------------|----------------------------------------------------------------------------------------------|
| Documented quality management system | Lab has controlled procedures for every calibration they perform                             |
| Measurement traceability to SI       | Every measurement is traceable to national/international standards through an unbroken chain |
| Measurement uncertainty evaluation   | Lab quantifies and reports how much uncertainty exists in their measurements                 |
| Proficiency testing participation    | Lab regularly compares results with peer labs to verify competence                           |

| Requirement                                   | What It Means for You                                                                     |
|-----------------------------------------------|-------------------------------------------------------------------------------------------|
| Competent personnel                           | Technicians are trained, qualified, and supervised per documented procedures              |
| Appropriate environment                       | Lab maintains controlled temperature, humidity, and cleanliness for precision measurement |
| Equipment calibrated by higher-tier standards | Lab's own reference standards are calibrated by national or primary-tier labs             |

### 2025-2026 Update

The 2025 edition of ISO/IEC 17025 now explicitly requires management of digital calibration data, electronic signatures, and validation of software used in measurement or reporting — including support for Digital Calibration Certificates (DCCs). Verify that your lab is current with these requirements.

# Scope of Accreditation — The Most Important Document

## Always Check the Scope — Not Just the Certificate

An accreditation certificate confirms a lab is accredited. The **scope of accreditation** — a separate, detailed document — specifies exactly what they're accredited to calibrate, including parameter ranges, best measurement capability (BMC), and calibration methods. A lab accredited for dimensional calibration is NOT automatically accredited for pressure or temperature calibration.

## Sample scope entry

| Parameter                  | Range             | Best Measurement Capability (BMC) | Method                              |
|----------------------------|-------------------|-----------------------------------|-------------------------------------|
| DC Voltage                 | 0 to 1,000 V      | 0.0018% of reading + 0.5 $\mu$ V  | Fluke 5730A per lab procedure       |
| Pressure (gauge)           | 0 to 10,000 psi   | 0.01% of reading                  | Deadweight tester per lab procedure |
| Length (gage blocks)       | 0.01" to 20"      | 2 $\mu$ in + 0.5 $\mu$ in/in      | Comparison per ASME B89.1.9         |
| Temperature (thermocouple) | -200°C to 1,200°C | 0.3°C to 1.5°C (range-dependent)  | Comparison per lab procedure        |

## How to Read a Scope of Accreditation

- 1 Parameter** — What measurement quantity (voltage, pressure, length, temperature)
- 2 Range** — The measurement range the lab can calibrate within
- 3 BMC** — Best Measurement Capability: the smallest uncertainty the lab can achieve under ideal conditions. Your actual uncertainty will be larger.
- 4 Method** — The reference standard and procedure used

**The #1 mistake in selecting a calibration lab**

is verifying they're accredited without checking their scope. A lab accredited for electrical calibration cannot provide accredited pressure calibration — even if they offer the service. Work performed outside the scope of accreditation is not accredited work.

A2LA, NVLAP, and ANAB all publish searchable scope-of-accreditation databases on their websites. Before contacting a lab, search their scope online to confirm they're accredited for the specific parameters and ranges you need.

# Industry-Specific Requirements

## Which accreditation and quality requirements apply to your industry?

| Industry                         | Calibration Requirement                             | Key Standard                  | What Your Auditor Checks                                                       |
|----------------------------------|-----------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------|
| Aerospace (AS9100)               | 17025-accredited calibration required for all M&TE  | AS9100 Rev D, Clause 7.1.5    | Scope covers your parameters; certificates match instruments                   |
| Aerospace (Nadcap)               | 17025-accredited; lab may need Nadcap accreditation | AC7000 series                 | Lab accreditation, scope, uncertainty ratios, proficiency testing              |
| Pharmaceutical (FDA/cGMP)        | "Suitable" calibration per 21 CFR 211/820           | 21 CFR 211.68, 820.72         | Traceability, calibration intervals, OOT procedures                            |
| Automotive (IATF 16949)          | 17025-accredited external calibration               | IATF 16949:2016, Clause 7.1.5 | Lab accreditation, MSA studies, scope of accreditation                         |
| Nuclear (10 CFR 50)              | 17025-accredited or NQA-1 supplier                  | 10 CFR 50 Appendix B          | Lab audit, traceability, quality program                                       |
| Defense (MIL-STD / ANSI Z540.3)  | ANSI/NCSL Z540.3 compliant calibration              | Z540.3-2006                   | 2% decision risk rule; lab must calculate and report measurement decision risk |
| General manufacturing (ISO 9001) | "Traceable" calibration — 17025 strongly preferred  | ISO 9001:2015, 7.1.5          | Traceability chain, calibration status of M&TE                                 |
| Environmental testing            | 17025-accredited for regulatory methods             | EPA, state regulations        | Lab accreditation for specific test methods                                    |

### ANSI/NCSL Z540.3 (Defense/Government)

This standard requires that the calibration provider limit the probability of false-accept decisions to no more than 2% by default. Your lab must calculate measurement decision risk — not just report pass/fail. If your customer or contract references Z540.3, your lab must comply with the decision risk requirement.

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# How to Verify Lab Accreditation

## Step-by-step verification process

Ask the lab for their accreditation certificate AND scope of accreditation. The certificate tells you they're accredited; the scope tells you for what. Both are required.

Verify the accreditation is current. Check the expiration date on the certificate. Accreditations are typically valid for 2–4 years with annual surveillance audits. A lapsed accreditation is no accreditation.

Verify the scope covers your specific calibration needs. Check: (a) the measurement parameter, (b) the range, and (c) the BMC is adequate for your required test uncertainty ratio (TUR).

Cross-reference using the accreditation body's online database:

- **A2LA:** Search at [a2la.org/directory](https://a2la.org/directory)
- **NVLAP:** Search at [nist.gov/nvlap](https://nist.gov/nvlap)
- **ANAB:** Search at [anab.org/directory](https://anab.org/directory)
- **PJLA:** Search at [pjlabs.com](https://pjlabs.com)

Check for the ILAC MRA mark on certificates. This mark confirms the accreditation is recognized internationally under the mutual recognition arrangement.

Ask about proficiency testing (PT) participation and results. Accredited labs must participate in inter-laboratory comparison programs. Consistent PT failures or non-participation are serious concerns.

If your industry requires it (Nadcap, nuclear), request the lab's quality manual and conduct a supplier audit. Accreditation is a baseline — your specific quality requirements may exceed it.

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# 10 Red Flags of an Unqualified Calibration Lab

1

## **Claims accreditation but can't produce a scope of accreditation.**

The certificate alone is insufficient. The scope document defines exactly what the lab is accredited to calibrate — without it, you cannot verify capability.

2

## **Scope doesn't cover your specific parameters.**

A lab accredited for DC voltage calibration is not accredited for RF power measurements. Scope matters — check parameter, range, and BMC.

3

## **Cannot state measurement uncertainty.**

Every accredited lab must be able to report measurement uncertainty per the GUM (Guide to the Expression of Uncertainty in Measurement). If they can't tell you their uncertainty for your measurement, their results are meaningless.

4

## **No environmental controls in the calibration area.**

Precision calibration requires controlled temperature (typically  $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$  for dimensional,  $23^{\circ}\text{C} \pm 2^{\circ}\text{C}$  for electrical), humidity, and vibration isolation. A lab calibrating micrometers in an uncontrolled warehouse is producing junk data.

5

## **Calibration certificates lack traceability information.**

Every certificate must identify the reference standards used, their calibration status, and the traceability chain to SI (typically through NIST or another NMI). No traceability = no value.

6

## **No proficiency testing participation.**

Accredited labs must participate in inter-laboratory comparisons (ILCs or proficiency testing programs). Refusal to share PT results or non-participation is a major concern.

7

## **Turnaround too fast to be credible.**

If a lab promises same-day turnaround on 50 instruments at the lowest price, they're likely skipping measurement points, not performing adjustments properly, or rubber-stamping certificates.

8

## **No out-of-tolerance notification process.**

When a reference standard or customer instrument is found out of tolerance, the lab must notify affected customers. Ask about their OOT notification procedure. If they don't have one, prior calibrations may be invalid.

**9****Pressure to skip "as-found" data.**

As-found data (the instrument's readings before adjustment) is essential for evaluating instrument drift and determining if previous measurements were valid. Labs that skip or charge extra for as-found data are cutting a critical corner.

**10****Staff cannot explain their procedures.**

A qualified lab's technicians can explain the calibration method, reference standard, uncertainty contributors, and acceptance criteria for any instrument in their scope. Inability to explain signals undertrained staff.

# Test Uncertainty Ratio (TUR) — A Buyer's Guide

## Why TUR Matters When Choosing a Lab

Test Uncertainty Ratio (TUR) is the ratio of the instrument's tolerance to the lab's measurement uncertainty. A TUR of 4:1 has been the traditional minimum — meaning the lab's uncertainty is at least 4 times smaller than the tolerance you're trying to verify. Higher TUR means more confidence in the calibration result.

## TUR Calculation

$$\text{TUR} = \text{Instrument Tolerance} / \text{Lab Measurement Uncertainty}$$

| TUR           | Confidence Level                                        | When Acceptable                                           |
|---------------|---------------------------------------------------------|-----------------------------------------------------------|
| 4:1 or higher | High confidence (~97% confidence in pass/fail decision) | Standard for most industrial calibration                  |
| 3:1           | Moderate confidence                                     | Acceptable per ANSI Z540.3 with guard-banding             |
| 2:1           | Low confidence — significant false-accept risk          | Generally unacceptable; requires risk analysis per Z540.3 |
| 1:1           | Unacceptable — uncertainty equals tolerance             | Calibration result is meaningless                         |

## Example

- Your pressure gauge has a tolerance of  $\pm 1$  psi
- Lab A's BMC for that range is  $\pm 0.1$  psi  $\rightarrow$  TUR = 10:1 (excellent)
- Lab B's BMC for that range is  $\pm 0.5$  psi  $\rightarrow$  TUR = 2:1 (unacceptable)
- Lab A's accredited calibration is far more valuable than Lab B's, even if Lab B is cheaper

### Cheaper is not better if the lab can't achieve adequate TUR.

A calibration certificate from a lab with poor TUR gives you a false sense of compliance. Always compare the lab's BMC (from their scope of accreditation) against your instrument's tolerance before selecting a provider.

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# Sample Vetting Questionnaire

## Calibration Lab Vetting Questionnaire — Send This Before Awarding Work

*Copy and send these questions to any calibration lab you're evaluating. Their answers will tell you everything you need to know about their capability and quality.*

### 1. Accreditation & Scope

- Are you ISO/IEC 17025 accredited? By which accreditation body (A2LA, NVLAP, ANAB, other)?
- Can you provide your current scope of accreditation?
- Does your scope cover the specific parameters, ranges, and BMCs we need? (Attach instrument list)
- When is your next surveillance audit? When does your accreditation expire?

### 2. Technical Capability

- What reference standards do you use for [our specific calibration needs]?
- What is your measurement uncertainty (BMC) for [our specific parameters and ranges]?
- What TUR can you achieve against our instrument tolerances?
- Do you comply with ANSI/NCSL Z540.3 (measurement decision risk)?

### 3. Quality System

- Do you participate in proficiency testing programs? Can you share recent results?
- What is your out-of-tolerance notification procedure?
- Do you report as-found and as-left data on all calibrations?
- What is your procedure for handling instruments found out of tolerance?
- How do you handle nonconformances in your quality system?

### 4. Logistics & Service

- What is your standard turnaround time? Rush turnaround and cost?
- Do you offer on-site calibration services? What is the day rate?
- Do you offer pickup/delivery service?
- What is your procedure for receiving, handling, and storing customer instruments?
- What software system do you use for managing calibration records and certificates?

## **5. Staffing & Environment**

- How many calibration technicians do you employ? What are their qualifications?
- What are your lab environmental conditions (temperature, humidity, vibration)?
- Do you maintain environmental monitoring records?

## **6. Pricing & Terms**

- Can you provide a fixed per-instrument price list for our instrument types?
- What additional charges apply? (Rush, repair, adjustment, shipping, certs, labels)
- Do you offer volume discounts or annual contract pricing?
- What is your payment terms and cancellation policy?

# CalLedger

## CalLedger

The most comprehensive directory of accredited calibration laboratories in the United States

**Search labs at [calledger.com](https://calledger.com)**